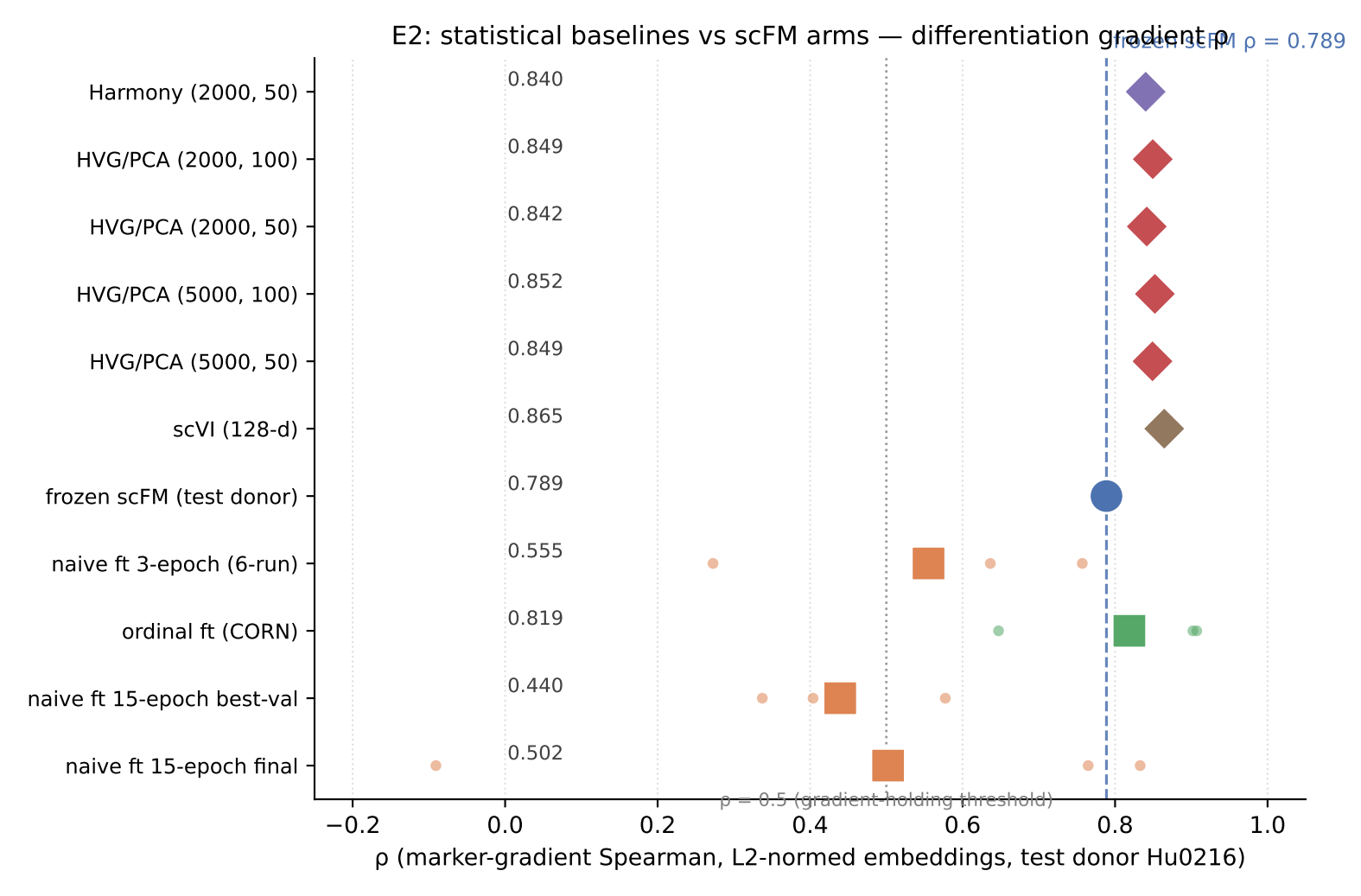
E2: statistical baselines vs scFM arms — differentiation gradient ρ
frozen scFM ρ = 0.789
Harmony (2000, 50)
HVG/PCA (2000, 100)
HVG/PCA (2000, 50)
HVG/PCA (5000, 100)
HVG/PCA (5000, 50)
scVI (128-d)
frozen scFM (test donor)
naive ft 3-epoch (6-run)
ordinal ft (CORN)
naive ft 15-epoch best-val
naive ft 15-epoch final
0.840
0.849
0.842
0.852
0.849
0.865
0.789
0.555
0.819
0.440
0.502
ρ = 0.5 (gradient-holding threshold)
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
ρ (marker-gradient Spearman, L2-normed embeddings, test donor Hu0216)